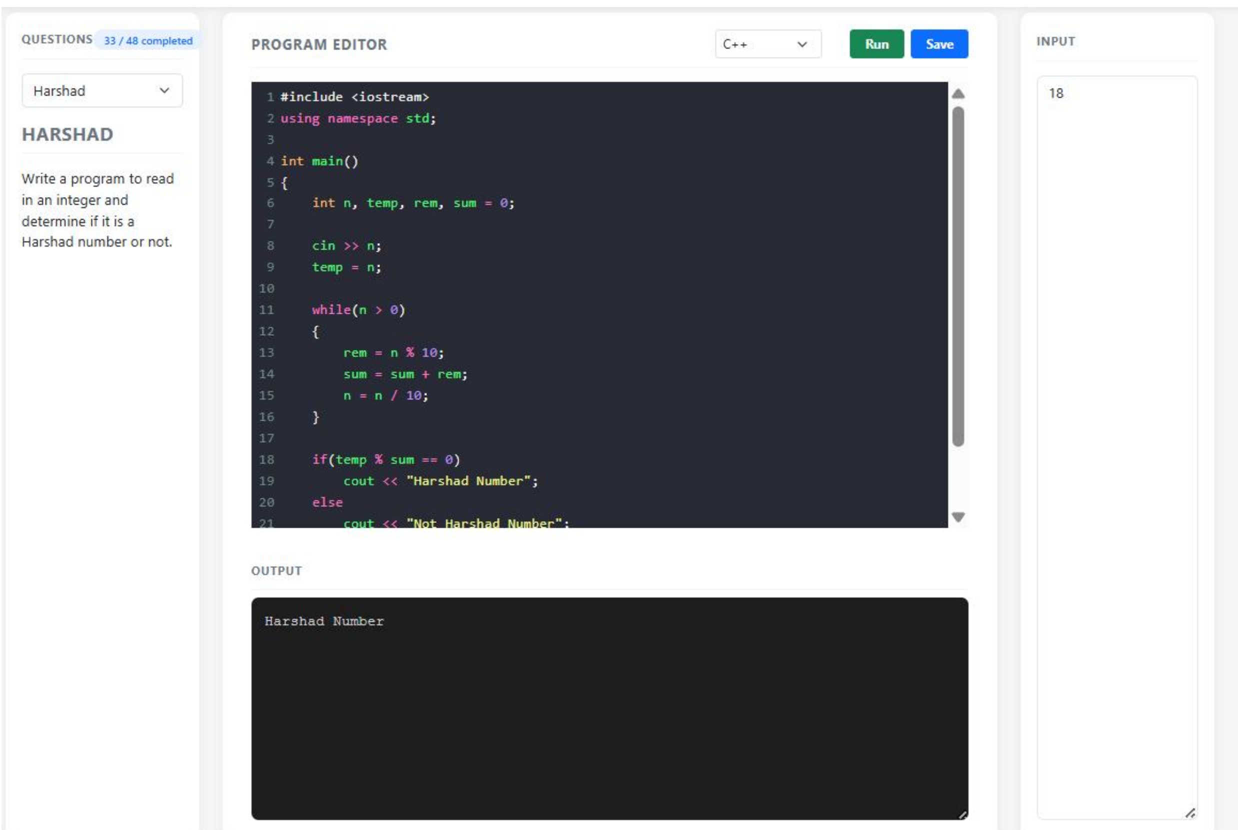
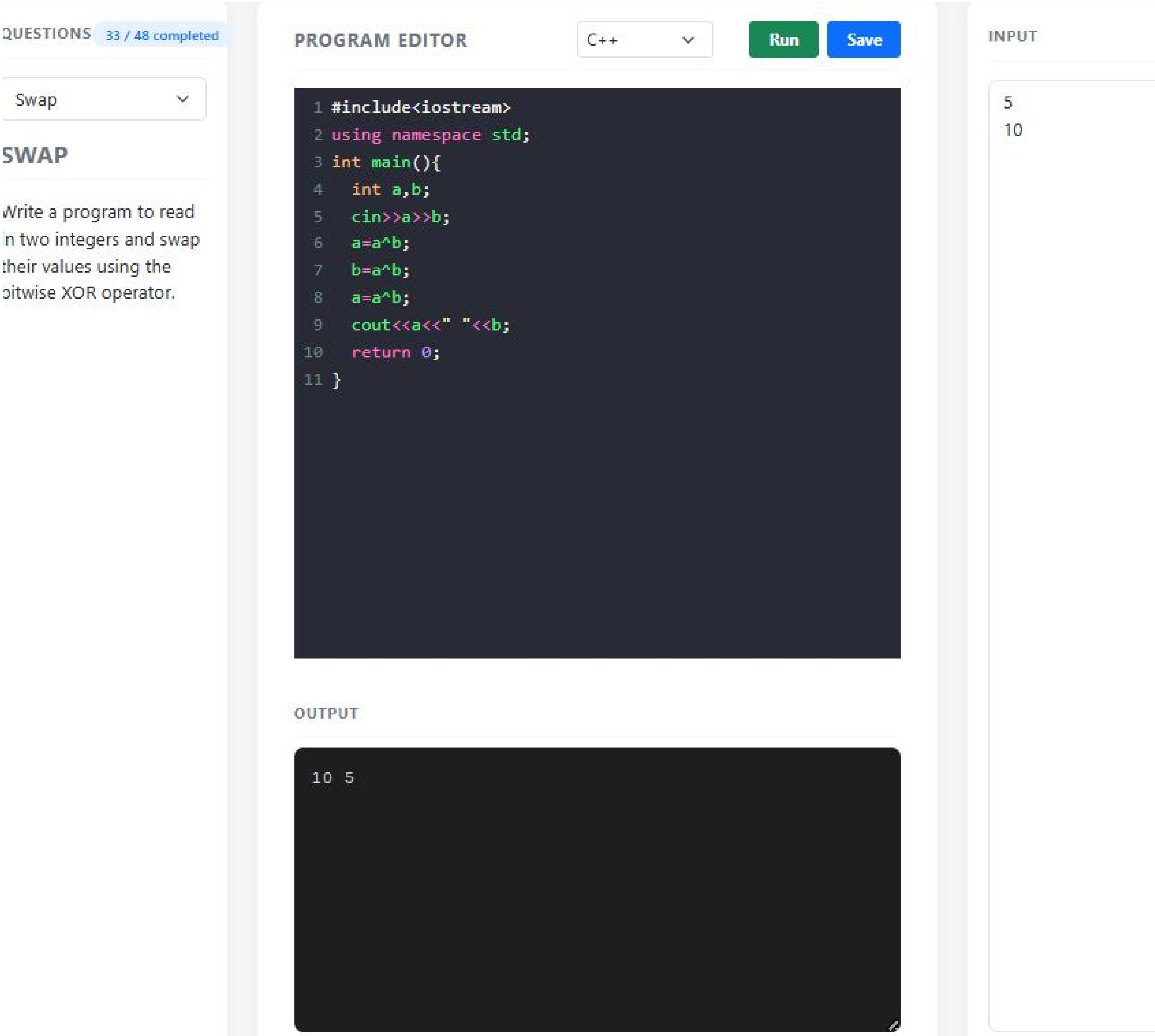
SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
COMPUTER SCIENCE AND ENGINEERING PROGRAMME
DSA0115 Object Oriented Programming with C++ for System Level Programming

LEVEL 1 QUESTIONS

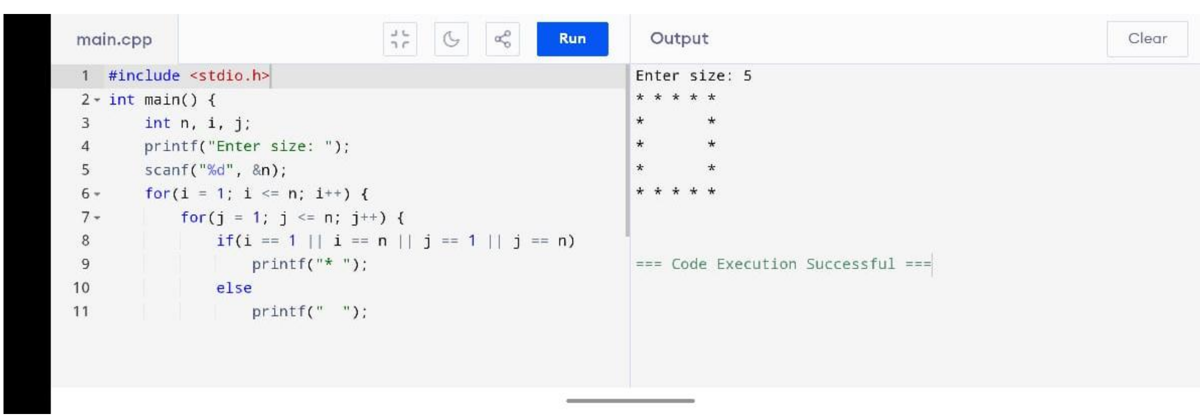
1. Write a program to check whether a number is a **Harshad number** (divisible by the sum of its digits).



2. Swap two numbers **without using a temporary variable**.



3. Hollow square pattern.



4. Number pattern.

Programiz C++ Online Compiler

Programiz PRO ❯

main.cpp

Run

```cpp
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter number of rows: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Output

Clear

```
Enter number of rows: 6
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6


=== Code Execution Successful ===
```

5. Check whether a number is a **palindrome**.

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Palindrome

PALINDROME

Write a program to read in an integer and determine if it is a palindrome or not using a while loop.

PROGRAM EDITOR

C++

Run Save

```cpp
#include <iostream>
using namespace std;
int main()
{
    int n, temp, rev = 0, rem;
    cin >> n;
    temp = n;
    while(n > 0)
    {
        rem = n % 10;
        rev = rev * 10 + rem;
        n = n / 10;
    }
    if(temp == rev)
        cout << "Palindrome";
    else
        cout << "Not Palindrome";
    return 0;
}
```

INPUT

101

OUTPUT

```
Palindrome
```

6. Find the **sum of the cubes of the first n natural numbers**.

main.cpp

```cpp
#include <iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        sum += i * i * i;
    }
    cout << "Sum of cubes = " << sum;
    return 0;
}
```

Output

```
Enter n: 10
Sum of cubes = 3025

=== Code Execution Successful ===
```

7. Convert **binary to decimal**.

main.cpp

```cpp
#include <iostream>
using namespace std;
int main() {
    int binary, decimal = 0, base = 1, rem;
    cout << "Enter a binary number: ";
    cin >> binary;
    while (binary > 0) {
        rem = binary % 10;
        decimal += rem * base;
        binary /= 10;
        base *= 2;  }
    cout << "Decimal = " << decimal;
    return 0;
}
```

Output

```
Enter a binary number: 1011
Decimal = 11

=== Code Execution Successful ===
```

8. Palindrome pattern.

main.cpp

```cpp
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter number of rows: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        for (int j = i; j >= 1; j--)
            cout << j;
        for (int j = 2; j <= i; j++)
            cout << j;
        cout << endl;    }
    return 0;
}
```

Output

```
Enter number of rows: 6
1
212
32123
4321234
543212345
65432123456

=== Code Execution Successful ===
```

9. Hollow pyramid pattern.

main.cpp

```cpp
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter number of rows: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n - i; j++)
            cout << " ";
        for (int j = 1; j <= 2 * i - 1; j++) {
            if (i == n || j == 1 || j == 2 * i - 1)
                cout << "*";
            else
                cout << " "; }
```

Output

```
Enter number of rows: 6
     *
    * *
   *   *
  *     *
 *       *
***********


=== Code Execution Successful ===
```

10. Find the **sum of the squares of the first n natural numbers**.

main.cpp

```cpp
#include <iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        sum += i * i;
    }
    cout << "Sum of squares = " << sum;
    return 0;
}
```

Output

```
Enter n: 8
Sum of squares = 204

=== Code Execution Successful ===
```